

Machine learning Aided Optimisation of Perovskite Solar Cells

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Abstract

This work presents a combined analytical and machine learning-based modeling framework for a trilayer all-inorganic perovskite solar cell (PSC). It includes $\text{Cs}_2\text{AgBiBr}_6$ (2.02 eV), Cs_2TiBr_6 (1.8 eV), and Cs_2SnI_6 (1.48 eV) as the active absorber layers. The device is structured as ITO/ETL/trilayer/HTL/anode, which aims to improve spectral absorption and carrier extraction. The model for carrier transport operates under both dark and illuminated conditions. It employs drift-diffusion equations, Poisson's equation, and continuity relations while considering recombination processes like Shockley–Read–Hall and radiative mechanisms. Generation rates are calculated using the Beer–Lambert law and integrated across the multilayer profile.

At the same time, a dataset of 700 simulated device instances is created using this physical model and saved in a PKL file. Five machine learning models, namely Linear Regression, XGBoost, CatBoost, Artificial Neural Network (ANN), and an Ensemble model, are trained to predict key photovoltaic parameters. These include open-circuit voltage (Voc), short-circuit current density (Jsc), fill factor (FF), and power conversion efficiency (PCE). Model evaluation shows that the Ensemble method achieves the highest predictive accuracy. This confirms the value of combining physical insights with data-driven learning. This dual approach serves as a strong tool for optimizing multilayer PSC design and predicting device performance under different configurations.

1. Introduction

What are Solar Cells?

Solar cells, also called photovoltaic (PV) cells, are devices that convert sunlight directly into electricity using the photovoltaic effect. They are the basic components of solar panels and are essential for capturing renewable energy from the sun.

Working Principle of Solar Cells:

The main function of a solar cell relies on the photovoltaic effect, which Alexandre-Edmond Becquerel discovered in 1839:

1. **Photon Absorption:** Sunlight hits the solar cell's surface. Photons with enough energy get absorbed by the semiconductor.
2. **Electron-Hole Pair Generation:** The absorbed energy causes electrons to move from the valence band to the conduction band, creating electron-hole pairs.

3. Charge Separation: An electric field at the p-n junction or heterojunction pushes electrons to the n-side and holes to the p-side.

4. Current Generation: This charge movement produces a current that can be drawn from external terminals, resulting in usable direct current (DC) electricity.

Types of Solar Cells:

1. Crystalline Silicon (c-Si) Solar Cells

- Monocrystalline: High efficiency (20–25%), expensive.
- Polycrystalline: Moderate efficiency (15–20%), lower cost.

2. Thin-Film Solar Cells

- Amorphous Silicon (a-Si): Flexible, low efficiency.
- CdTe (Cadmium Telluride): Cheaper, moderate efficiency.
- CIGS (Copper Indium Gallium Selenide): High efficiency, complex to make.

3. Perovskite Solar Cells

- Based on ABX_3 perovskite structure.
- High tunability, low-cost production, but stability is a challenge.
- Includes organic-inorganic types (e.g., $MAPbI_3$) and all-inorganic types (e.g., $CsSnBr_3$, $CsPbI_3$).

4. Organic Photovoltaics (OPVs)

- Use conductive polymers or small molecules.
- Lightweight, flexible, low efficiency (~10–12%).

5. Quantum Dot and Dye-Sensitized Solar Cells

- Use nanomaterials or dyes to capture light.
- Promising for specific, low-light applications.

Key Parameters

- Voc: Open-circuit voltage

- Jsc: Short-circuit current density
- FF: Fill factor
- PCE: Power conversion efficiency

What is Lead-Free Perovskite Solar Cells?

Lead-free perovskite solar cells are a type of perovskite photovoltaics that remove toxic lead (Pb^{2+}) from the light-absorbing layer. They use safer elements like tin (Sn^{2+}), bismuth (Bi^{3+}), antimony (Sb^{3+}), silver (Ag^+), and copper (Cu^+), making them more environmentally friendly and sustainable.

Why Go Lead-Free?

Traditional perovskite solar cells (e.g., $\text{CH}_3\text{NH}_3\text{PbI}_3$) have shown impressive power conversion efficiencies (PCE > 25%), but the presence of toxic lead raises environmental and regulatory issues, especially for widespread use. Lead can contaminate water and soil if not carefully handled.

1. Common Lead-Free Perovskite Materials

- Tin-Based Perovskites (e.g., CsSnI_3 , FASnI_3)
- Closest to Pb-based perovskites.
- High mobility, but struggle with stability due to easy oxidation of Sn^{2+} to Sn^{4+} .

2. Double Perovskites (e.g., $\text{Cs}_2\text{AgBiBr}_6$)

- Replace two Pb^{2+} ions with one Ag^+ and one Bi^{3+} .
- Highly stable, but often have indirect bandgaps, lowering efficiency.

3. Bismuth-Based Compounds (e.g., $\text{MA}_3\text{Bi}_2\text{I}_9$)

- Non-toxic and chemically stable.
- Lower efficiency due to poor charge movement.

4. Antimony-Based Compounds (e.g., $\text{Cs}_3\text{Sb}_2\text{I}_9$)

- Similar to Bi-based perovskites but with adjustable properties.
- Still under active research.

Advantages

- Non-toxic and environmentally friendly.
- Potential for thermal and chemical stability.
- Fit for low-cost, solution-based production methods.

Challenges

- Lower power conversion efficiency (PCE often < 15%).
- Material instability, especially with Sn-based perovskites.
- Defect tolerance and charge carrier movement are usually worse than lead-based options.

Current Research Directions

- Passivation methods to stabilize Sn^{2+} oxidation.
- Compositional changes (e.g., mixing Sn with Ge or Bi).
- Layered device designs like trilayer or graded bandgap structures.
- Use machine learning and physics-based models for predicting performance and optimization.

2.Literature survey:

Paper 1: Optical properties and electronic structures of lithium doped double perovskite $\text{Cs}_2\text{AgBiCl}_6$ crystals

This study looks at the effects of Li^+ doping in lead-free double perovskite $\text{Cs}_2\text{AgBiCl}_6$ crystals. The goal is to improve optical properties for optoelectronic uses while avoiding the toxicity of lead-based perovskites.[2]

Key Contributions:

- Synthesis: $\text{Cs}_2\text{Li}_x\text{Ag}_{1-x}\text{BiCl}_6$ ($x = 0-1$) crystals were grown using a heat-assisted solution evaporation method.[3]
- Structural Analysis: XRD confirms that the double perovskite cubic structure (Fm-3m) remains intact. Rietveld refinements show lattice contraction due to Li^+ doping.
- Morphology: SEM and HRTEM reveal smaller grains and a tighter lattice in Li-doped samples.
- Optical Properties: Li^+ doping leads to:

1. A blue shift in the absorption edge.
2. Improved photoluminescence (PL) with broad emission from self-trapped excitons (STEs).
3. An increase in the band gap from 2.68 eV to 3.21 eV ($x=0$ to 1).

• DFT Calculations: Using PBE and HSE06, band structures and partial DOS show:

1. The indirect band gap is preserved.
2. The band gap increases with more Li.
3. Li does not contribute to the VBM or CBM but causes sublattice distortion, breaking parity-forbidden transitions.

Li^+ doping in $\text{Cs}_2\text{AgBiCl}_6$ improves photoluminescence by changing the electronic structure and disrupting symmetry-driven forbidden transitions. This increases its potential for use in LEDs and optoelectronics, particularly under harsh UV conditions.

Paper 2: Dimensional reduction in $\text{Cs}_2\text{AgBiBr}_6$ enables long-term stable perovskite-based gas sensing

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To address environmental instability in lead-free halide perovskite $\text{Cs}_2\text{AgBiBr}_6$ gas sensors, we reduced dimensions by substituting organic cations. This process created 2D hybrid double perovskites (HDPs) with strong CO gas sensing capabilities and lasting durability.[4]

Materials & Methodology

New 2D perovskites synthesized:

- a. $(\text{BA})_4\text{AgBiBr}_8$ (n-butylamine)
- b. $(\text{HA})_4\text{AgBiBr}_8$ (n-hexylamine)
- c. $(\text{OA})_4\text{AgBiBr}_8$ (n-octylamine)

Synthesis Strategy:

We replaced inorganic Cs^+ with long-chain alkylammonium cations (BA^+ , HA^+ , OA^+) to create layered 2D perovskites with better moisture and structural stability.

Characterization Techniques:

- a. SEM, HRTEM, PXRD, ATR-FTIR, Raman, UV-Vis
- b. DFT calculations, PDOS, SFG spectroscopy, ELF, COHP

Key Results

1. Mechanism Insight:

- a. CO adsorbs physically at Bi-defect sites, which causes localized charge redistribution.
- b. DFT and PDOS confirm weak orbital hybridization and strong charge transfer.
- c. SFG spectroscopy shows CO alignment on the perovskite surface after exposure.

This work represents a major breakthrough in CO gas sensing using environmentally stable 2D lead-free perovskites, specifically $(\text{BA})_4\text{AgBiBr}_8$. It:

- a. Detects CO at levels as low as 20 ppb
- b. Maintains structural and sensing integrity for over 9 months
- c. Outperforms traditional sensors and earlier perovskite-based sensors

Paper 3: Breakthrough in sustainable photovoltaics: Achieving 30.86 % efficiency with innovative lead-free bilayer perovskite solar cells using SCAPS-1D and DFT framework.

1. Objectives:

- a. Develop high-efficiency and eco-friendly bilayer PSCs with CsSnBr_3 and CsSnI_2Br .
- b. Use DFT and SCAPS-1D simulations to improve structural, optical, and electronic parameters.

2. Final Configuration:

ITO / SnS_2 / CsSnBr_3 / CsSnI_2Br / CBTS / C

- a. CsSnBr_3 : Top absorber (high-energy photons, more stable)
- b. CsSnI_2Br : Bottom absorber (low-energy photons, higher absorption)
- c. SnS_2 : Best ETL
- d. CBTS: Best HTL (highest Voc and stability)

3. Simulation Highlights:

- Single-junction baseline PCEs: 14.15% (CsSnBr_3), 14.59% (CsSnI_2Br)
- Bilayer optimized PCE: 30.86%
- Absorber optimization: CsSnBr_3 at 1.5 μm , CsSnI_2Br at 1.0 μm
- Best HTL: CBTS; offers 15.47% PCE and the best Voc (0.9658 V)
- Best ETL: SnS_2 ; yields the highest PCE (16.51%) among alternatives

4. Key Equations Used:

- Poisson's equation
- Continuity equations for electrons and holes
- Drift-diffusion transport equations
- Shockley-Read-Hall recombination model

The study successfully shows a lead-free, all-inorganic bilayer PSC with much better performance. The 30.86% PCE from the $\text{CsSnBr}_3/\text{CsSnI}_2\text{Br}$ tandem stack beats most previously reported structures, highlighting its potential for future solar technologies.

3. Unexplored Areas in Lead-Free Trilayer Perovskite Solar Cells

1. Optimized Material Combinations for Trilayer Architectures

- Most studies focus on single or bilayer structures. Trilayer combinations like $\text{CsSnBr}_3/\text{CsSnI}_2\text{Br}$ are still underexplored.
- Layer-wise band alignment engineering aims to minimize recombination at interfaces, but it mainly relies on trial-and-error.

2. Stability Enhancement Strategies for Sn-Based Perovskites

- The oxidation of Sn^{2+} to Sn^{4+} reduces device lifespan. We do not fully understand long-term air and moisture stability under real-world conditions.
- More research is needed on encapsulation, dopants, and additive passivation methods specific to Sn-based absorbers.

3. Advanced Interface Engineering

- Interfaces between $\text{SnS}_2/\text{CsSnBr}_3$ and $\text{CsSnI}_2\text{Br}/\text{CBTS}$ often face issues due to defect-induced recombination.
- Few studies focus on 2D/3D hybrid interlayers, buffer layers, or tuning band offsets to reduce interfacial losses in trilayer cells.

4. Device-Scale Simulation Using Coupled Electro-Optical Models

- Many models only solve electrical equations such as Poisson and continuity. We lack integrated optical, electrical, and thermal simulations for multilayer lead-free devices.

5. Data-Driven Inverse Design Using ML

- While ML is used for performance prediction, its application for inverse design (such as predicting layer thickness, bandgap, or doping based on target PCE) is still emerging.

6. Limited Experimental Validation for Theoretical Models

- Most theoretical studies do not have experimental validation because it is difficult to synthesize and characterize specific layered materials like CsSnI_2Br .

7. Lead-Free Tandem Configurations

- Trilayer PSCs show promise for tandem cells, but there have been few attempts to combine them with Si or CIGS bottom cells in a lead-free tandem setup.

8. Scalable Fabrication Techniques

- Spin coating is commonly used for lab-scale devices. Scalable methods like blade coating, spray pyrolysis, or CVD for trilayer structures remain largely untested.

4. Motivation and Objectives

Motivation

The rapid increase in global energy demand, along with growing environmental concerns, highlights the need for clean and sustainable energy solutions. Perovskite solar cells (PSCs) have become one of the most promising solar technologies. They have high power conversion efficiency (PCE), low fabrication costs, and adjustable optoelectronic properties. However, traditional PSCs often use lead-based compounds, which present serious environmental and health risks. This has sparked interest in lead-free perovskite materials like CsSnBr_3 and CsSnI_2Br , which provide a non-toxic and stable alternative.

Additionally, traditional modeling methods for solar cell design usually depend on numerical simulation or trial-and-error experiments, both of which can be time-consuming and resource-heavy. Combining physics-based mathematical modeling with machine learning (ML) allows for quicker and more accurate predictions of device performance. It also supports the intelligent optimization of materials and structures. In this context, using a trilayer graded bandgap architecture can improve light absorption and carrier transport. However, this approach remains largely unexamined in lead-free PSCs, presenting a significant opportunity for increasing efficiency.

Objectives

This research aims to:

1. Design and model a lead-free tri-layer perovskite solar cell structure using $\text{Cs}_2\text{AgBiBr}_6$, Cs_2TiBr_6 and Cs_2SnI_6 as absorber layers, along with TiO_2 (ETL) and Cu_2O (HTL).
2. Develop a model to analyze carrier transport under both dark and illuminated conditions using Poisson's equation, drift-diffusion, and continuity equations. This will consider radiative and SRH recombination.
3. Simulate the generation profile and current density-voltage (J–V) characteristics using Beer–Lambert law and field-assisted drift-diffusion transport mechanisms.
4. Generate a dataset of 700 simulation points from the model to train and test various machine learning models.
5. Implement and compare five ML algorithms: Linear Regression, XGBoost, CatBoost, Artificial Neural Network (ANN), and an Ensemble model. These will be used to predict the performance of key solar cell parameters (V_{oc} , J_{sc} , FF, and PCE).

6. Identify the most accurate ML model and interpret how physical and geometric parameters influence solar cell performance.
7. Explore the potential of the combined approach of analytical methods and machine learning for future lead-free, high-efficiency PSC design and optimization.

5. Methodology

In this work, we conduct a detailed modeling and performance analysis for a lead-free tri-layer perovskite Solar cell (PSC) using both analytical equations and machine learning methods. The proposed structure includes **TiO₂** as the electron transport layer (ETL), **Cs₂AgBiBr₆**, **Cs₂TiBr₆** and **Cs₂SnI₆** as triple absorber layers, and **Cu₂O** as the hole transport layer (HTL); this provides a fully inorganic, non-toxic alternative to standard lead-based PSCs. We perform thorough mathematical modeling under both dark and illuminated conditions, using drift-diffusion, Poisson's, and continuity equations while considering radiative and Shockley-Read-Hall (SRH) recombination. We calculate the electric field distribution, carrier concentration profiles, and generation rate for each layer. Meanwhile, we generate a dataset of 700 simulated devices based on this physics-based model. This dataset is used to train five machine learning models: Linear Regression, XGBoost, CatBoost, Artificial Neural Network (ANN), and an Ensemble model. These models predict key performance metrics such as Voc, Jsc, FF, and PCE, with the ensemble approach achieving the best accuracy. This dual approach gives both physical insight and data-based optimization, allowing for quick screening and design of efficient, stable, lead-free PSC architectures:

Analytical modeling for a proposed tri-layer Perovskite Solar Cell

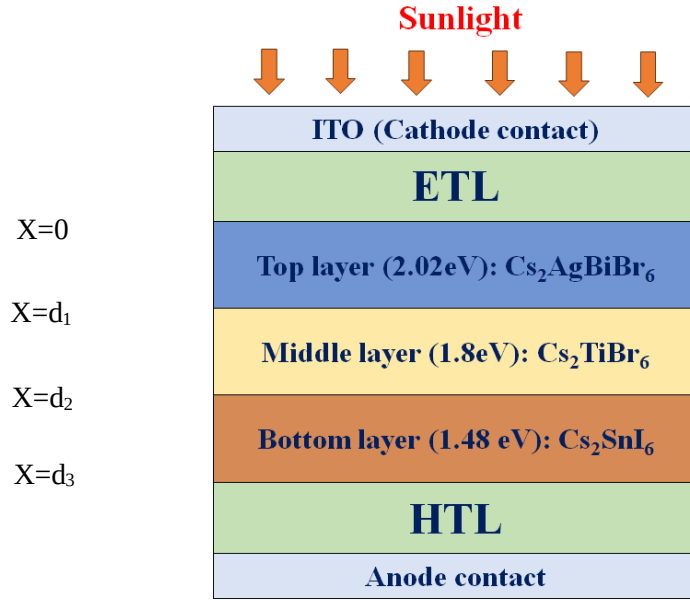


Fig.1 Schematic of the proposed tri-layer PSC structure

$$x = 0, x = d_1, d_2, d_3$$

Key assumptions:

1. No generation $G(x)=0$
2. Steady state $\left(\frac{\delta n}{\delta t}\right) = 0$
3. Constant Electric Field.
4. Radiative and non-radiative recombination.
5. ETL and HTL do not contribute to photo current.

For electrons in dark conditions:

$$\frac{dJ_n}{dx} = -qR(x) \tag{1}$$

$$\text{Where } J_n = q\mu_n n(x)E + qD_n \frac{dn(x)}{dx}$$

After substituting,

$$\frac{d}{dx} \left[q\mu_n n(x)E + qD_n \frac{dn(x)}{dx} \right] = -qR(x) \tag{2}$$

$$\mu_n E \frac{dn}{dx} + D_n \frac{d^2 n}{dx^2} = -R(x)$$

We know,

$$R_{SRH} = \frac{np - ni^2}{\tau_p(n + ni) + \tau_n(p + pi)}$$

$$R_{rad} = -B(np - ni^2)$$

where n = free electron concentration and p= free whole concentration

Approximation: $n=p$ & $n^2 \gg ni^2$

$$R(x) = \frac{n}{\tau_{SRH}} + Bn^2 \quad (3)$$

$$\frac{1}{\tau_{eff}} = \frac{1}{\tau_{srh}} + Bn \quad (4)$$

$$\Rightarrow R(x) = \frac{n(x)}{\tau_{eff}} \quad (5)$$

Thus, the continuity equation becomes:

$$D_n \frac{d^2 n}{dx^2} + \mu_n E \frac{dn}{dx} - \frac{n(x)}{\tau_{eff}} = 0 \quad (6)$$

The solution becomes,

$$n(x) = Ae^{a_1 x} + Be^{a_2 x} \quad (7)$$

Where a_1 and a_2 are the roots of the following quadratic equations.

$$a^2 + \frac{\mu_n E}{D_n} a - \frac{1}{D_n \tau_{eff}} = 0$$

Boundary conditions depend on ETL and HTL interfaces

Boundary Conditions:

- i. At ETL ($x = 0$), $n(0) = n_0 e^{\frac{qV}{kT}}$, V = anode voltage, K = Boltzmann constant, T = absolute temperature
- ii. At HTL ($x = d_3$), $n(d_3) = 0$, perfect extraction of holes.

Now applying these boundary conditions to find A&B.

Applying (i)& (ii) in equation (7),

$$n(0) = Ae^0 + Be^0$$

$$n_0 e^{\frac{qV}{KT}} = A + B$$

$$0 = Ae^{a_1 d_3} + Be^{a_2 d_3}$$

$$A = \frac{-Be^{a_2 d_3}}{e^{a_1 d_3}}$$

After substituting,

$$n_0 e^{\frac{qV}{KT}} = B - Be^{d_3(a_2 - a_1)}$$

$$\Rightarrow B(1 - e^{d_3(a_2 - a_1)}) = n_0 e^{\frac{qV}{KT}}$$

$$B = \frac{n_0 e^{\frac{qV}{KT}}}{1 - e^{d_3(a_2 - a_1)}}$$

$$A = \frac{n_0 e^{\frac{qV}{KT}}}{1 - e^{d_3(a_2 - a_1)}} e^{d_3(a_2 - a_1)}$$

$$\text{At } x = 0, J_{dark} = J_n(0)$$

$$J_n(x) = q\mu_n n(x)E + qD_n \frac{dn(x)}{dx}$$

$$J_n(0) = q\mu_n n(0)E + qD_n \frac{dn(0)}{dx}$$

$$n(0) = A + B$$

$$J_n(0) = q\mu_n n(0)E$$

Similarly, we can obtain $J_p(0) = q\mu_p p(0)E$

$$J_{dark} = J_n(0) + J_p(0)$$

Under illumination:

From Poisson's Equation, we can get electric-field which can be expressed as,:

$$\frac{d^2\psi(x)}{dx^2} = \frac{-q}{E(x)} [p(x) - n(x) + N_D^+(x) - N_A^-(x)] \quad (8)$$

Where $\psi(x)$ = electrostatic potential, $E(x)$ = electric field, $n(x)$ & $p(x)$ are the free excess electron & hole concentrations, N_D & N_A be the donor and acceptor doping concentrations of absorber layers. These equations can be implied to find the field for separately three absorber layers.

Assuming absorber layers are intrinsic,

$$\frac{d^2\psi}{dx^2} \approx 0, E(x) = \frac{V_{bi}-V}{(d_1+d_2+d_3)} \quad (9)$$

Drift Diffusion Equations:

- i. Electron current density [2][3],

$$J_n(x) = q\mu_n n(x)E + qD_n \frac{dn(x)}{dx} \quad (10)$$

- ii. Hole current density,

$$J_p(x) = q\mu_p p(x)E + qD_p \frac{dp(x)}{dx}$$

Now under steady state solution,

$$\frac{dJ_n}{dx} = qG(x) - qR(x) \quad (11)$$

$$\frac{dJ_p}{dx} = qG(x) - qR(x) \quad (12)$$

$$R(x) = R_{SRH} + R_{rad} = \frac{n(x)}{\tau_{SRH}} + Bn^2(x) \quad (13)$$

$$\frac{1}{\tau_{eff}} = \frac{1}{\tau_{SRH}} + Bn(x)$$

Boundary conditions:

- i. $p(0) = 0, n(0) = n_{ETL} = n_0 e^{\frac{qV}{kT}}$
ii. $n(d) = 0, p(d) = p_{HTL} = p_0 e^{\frac{qV}{kT}}$

Generation Profile:

Assuming an exponentially decaying absorption (Beer-Lambert Law)

$$G(x) = \varphi_0 \alpha e^{-\alpha x} \quad (14)$$

Where, φ_0 is the incident photon flux, α is the absorption coefficient, we apply the above noted expression for the three different perovskite layers for our proposed device to calculate the exact generation rate.

Generation Rate for three perovskite absorber layers is given below:

$$G_i = \sum_{i=1}^{d_3} \alpha_i \varphi_0 e^{-\alpha_i x} dx \quad (15)$$

$$= \varphi_0 (1 - e^{-\alpha d_i}) \quad (16)$$

Now we will find out the expression of J-V relation for three different perovskite layers.

For Top absorber layer:

So, the continuity equation becomes:

$$D_{n1} \frac{d^2 n_1(x)}{dx^2} + \mu_n E \frac{dn_1(x)}{dx} - \frac{n_1(x)}{\tau_{eff}} + G_1 = 0 \quad (17)$$

$$n_1(x) = A_1 e^{b_1 x} + B_1 e^{b_2 x} + n_1(0)$$

$$n_1(0) = G \tau_{eff}$$

The photo-current can be expressed as [1], if the electric field is strong,

$$J_{ph}^{top} = q G_0 L_{1eff} [1 - \exp\left(-\frac{d_1}{L_{1eff}}\right)] \quad (18)$$

After substituting the L_{1eff} from equations (13)

$$J_{ph}^{top} = q G_0 \sqrt{D_{n1} \left(\frac{1}{\tau_{1srh} + B_1 N_1} \right)} [1 - \exp\left(-\frac{d_1}{\sqrt{D_{n1} \left(\frac{1}{\tau_{1srh} + B_1 N_1} \right)}}\right)] \quad (19)$$

Similarly calculate J_{ph}^{mid} & J_{ph}^{bott} , the expressions for middle and bottom layers are,

$$J_{ph}^{middle} = qG_0 \sqrt{D_{n2} \left(\frac{1}{\tau_{2srh}} + B_2 N_2 \right)} \left[1 - \exp \left(\frac{-d_2}{\sqrt{D_{n2} \left(\frac{1}{\tau_{2srh}} + B_2 N_2 \right)}} \right) \right] \quad (20)$$

$$J_{ph}^{bottom} = qG_0 \sqrt{D_{n3} \left(\frac{1}{\tau_{3srh}} + B_3 N_3 \right)} \left[1 - \exp \left(\frac{-d_3}{\sqrt{D_{n3} \left(\frac{1}{\tau_{3srh}} + B_3 N_3 \right)}} \right) \right] \quad (21)$$

The effective photo current combining all the photo-currents we have,

$$J_{ph} = \int_{\lambda_1}^{\lambda_2} q [\min(J_{ph}^{top}, J_{ph}^{mid}, J_{ph}^{bott})] \quad (22)$$

$$J_{ph} = \int_{\lambda_1}^{\lambda_2} q \left[\min \left(G_0 \sqrt{D_{n1} \left(\frac{1}{\tau_{1srh}} + B_1 N_1 \right)} \left[1 - \exp \left(\frac{-d_1}{\sqrt{D_{n1} \left(\frac{1}{\tau_{1srh}} + B_1 N_1 \right)}} \right) \right], G_0 \sqrt{D_{n2} \left(\frac{1}{\tau_{2srh}} + B_2 N_2 \right)} \left[1 - \exp \left(\frac{-d_2}{\sqrt{D_{n2} \left(\frac{1}{\tau_{2srh}} + B_2 N_2 \right)}} \right) \right], G_0 \sqrt{D_{n3} \left(\frac{1}{\tau_{3srh}} + B_3 N_3 \right)} \left[1 - \exp \left(\frac{-d_3}{\sqrt{D_{n3} \left(\frac{1}{\tau_{3srh}} + B_3 N_3 \right)}} \right) \right] \right)$$

$$J_{total} = J_{dark} - J_{ph}$$

$$= J_{dark} - \int_{\lambda_1}^{\lambda_2} q \left[\min \left(G_0 \sqrt{D_{n1} \left(\frac{1}{\tau_{1srh}} + B_1 N_1 \right)} \left[1 - \exp \left(\frac{-d_1}{\sqrt{D_{n1} \left(\frac{1}{\tau_{1srh}} + B_1 N_1 \right)}} \right) \right], G_0 \sqrt{D_{n2} \left(\frac{1}{\tau_{2srh}} + B_2 N_2 \right)} \left[1 - \exp \left(\frac{-d_2}{\sqrt{D_{n2} \left(\frac{1}{\tau_{2srh}} + B_2 N_2 \right)}} \right) \right], G_0 \sqrt{D_{n3} \left(\frac{1}{\tau_{3srh}} + B_3 N_3 \right)} \left[1 - \exp \left(\frac{-d_3}{\sqrt{D_{n3} \left(\frac{1}{\tau_{3srh}} + B_3 N_3 \right)}} \right) \right] \right) \right] \quad (24)$$

TCAD Simulation For Tri-layer Perovskite Solar Cell.

Table 1: Material parameters of perovskite solar cells from experimental works [7]

Parameters	ITO	TiO ₂	Cs ₂ AgBiBr ₆	Cs ₂ TiBr ₆	Cs ₂ SnI ₆	Cu ₂ O
Thickness (μm)	0.5	0.03	0.06	1	1	0.05
Bandgap (eV)	3.5	3.5	2	1.85	1.48	2.2
Electron Affinity (eV)	4	4	3.8	3.8	4.2	3.4
Relative Permittivity	9	9	14	10	11	7.5
CB Effective DOS (cm^{-3})	2.2×10^{18}	2×10^{18}	1×10^{18}	1×10^{18}	1×10^{18}	2×10^{19}
VB Effective DOS (cm^{-3})	1.8×10^{18}	1.8×10^{19}	1×10^{19}	1×10^{19}	1×10^{19}	1×10^{19}
Electron Mobility ($\text{cm}^2/\text{V}\cdot\text{s}$)	20	20	5	5	10	200
Hole Mobility ($\text{cm}^2/\text{V}\cdot\text{s}$)	10	10	5	5	1	8600
Donor Density ND (cm^{-3})	1×10^{19}	9×10^{16}	1×10^{18}	1×10^{18}	0	0
Acceptor Density NA (cm^{-3})	0	0	0	0	0	0
$N_{\text{tb}}/\text{cm}^{-2}$	1×10^{15}	1×10^{10}	1×10^{12}	1×10^{10}	1×10^{10}	1×10^{10}
Absorption Model	SCAPS qn (hv-Eg)	SCAPS qn (hv-Eg)	SCAPS qn (hv-Eg)	SCAPS qn (hv-Eg)	SCAPS qn (hv-Eg)	SCAPS qn (hv-Eg)

Table 2: Interface simulation parameter[7].

Parameters	TiO ₂ / Cs ₂ AgBiBr ₆	Cs ₂ AgBiBr ₆ / Cs ₂ TiBr ₆	Cs ₂ TiBr ₆ / Cs ₂ SnI ₆	Cs ₂ SnI ₆ / Cu ₂ O
Defect type	Neutral	Neutral	Neutral	Neutral
Capture cross section (e-h)	1×10^{-17}	1×10^{-17}	1×10^{-17}	1×10^{-17}
Energy Distribution	Single	Single	Single	Single
Reference for defect energy level	Above highest E_v	Above highest E_v	Above highest E_v	Above highest E_v
Energy w.r.t reference (eV)	0.6	0.6	0.6	0.6
Total Density (cm²)	1×10^{10}	1×10^{10}	1×10^{10}	1×10^{10}

Machine Learning Modelling for a proposed Tri-layer Perovskite Solar Cell

Model Comparison:

Five models were trained using 700 simulated data points: Linear Regression, Random Forest, XGBoost, CatBoost, and Artificial Neural Network (ANN). We evaluated their performance with R^2 scores and Root Mean Square Error (RMSE).

Table 3: Model Comparison

Model	Train R^2	Test R^2	Train RMSE	Test RMSE
Linear Regression	0.642	0.665	1.5719	1.3275
Random Forest	0.902	0.783	0.8197	1.0690
XGBoost	0.998	0.766	0.1085	1.1096
CatBoost	0.975	0.764	0.4166	1.1153
ANN	0.815	0.745	1.1312	1.1590
Ensemble	0.973	0.789	0.428	1.052

Random Forest had the best balance between training and testing accuracy. XGBoost showed a low training error, but it had a slightly higher generalization error, which means it was likely overfitting.

Table 4: Predicted PCE Values

Model	Predicted PCE (%)
Linear Regression	3.797
Random Forest	20.522
XGBoost	18.839
CatBoost	5.378
ANN	19.463

Models like Random Forest and ANN predicted PCE values similar to the analytical model, about 20%. This shows they have strong learning ability. Linear Regression and CatBoost did not perform as well, suggesting they lack the capacity to model nonlinear interactions.

6. Results & Discussions

6.1.1: Model calibration:

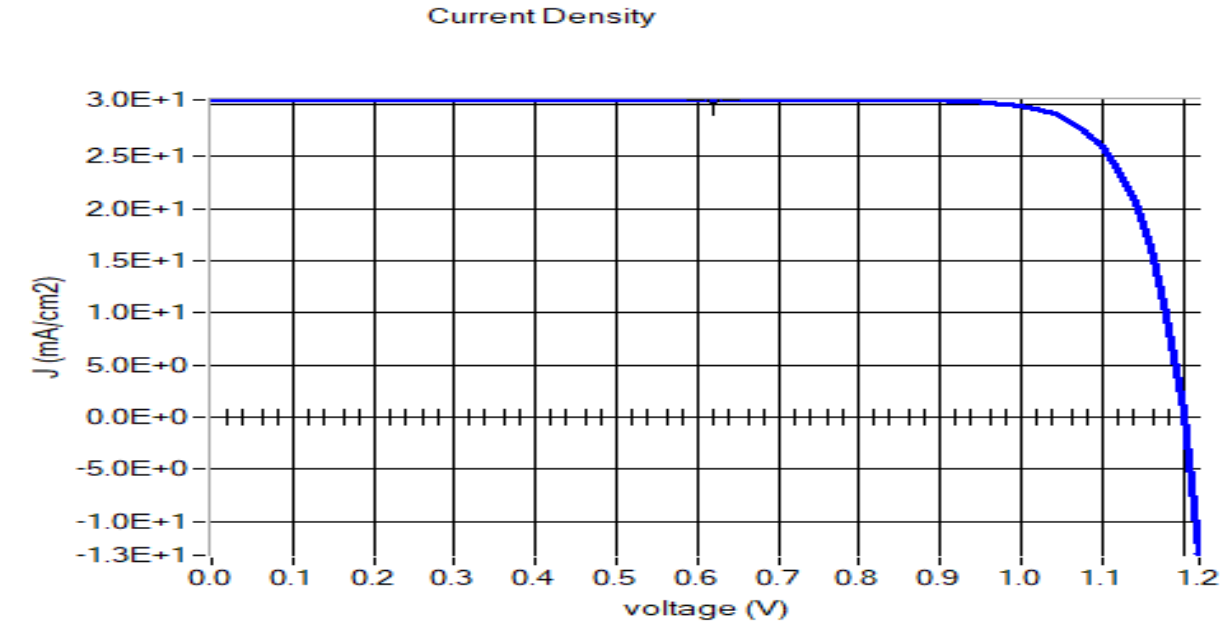


Fig.2 J-V characteristics graph of Bilayer PSC.

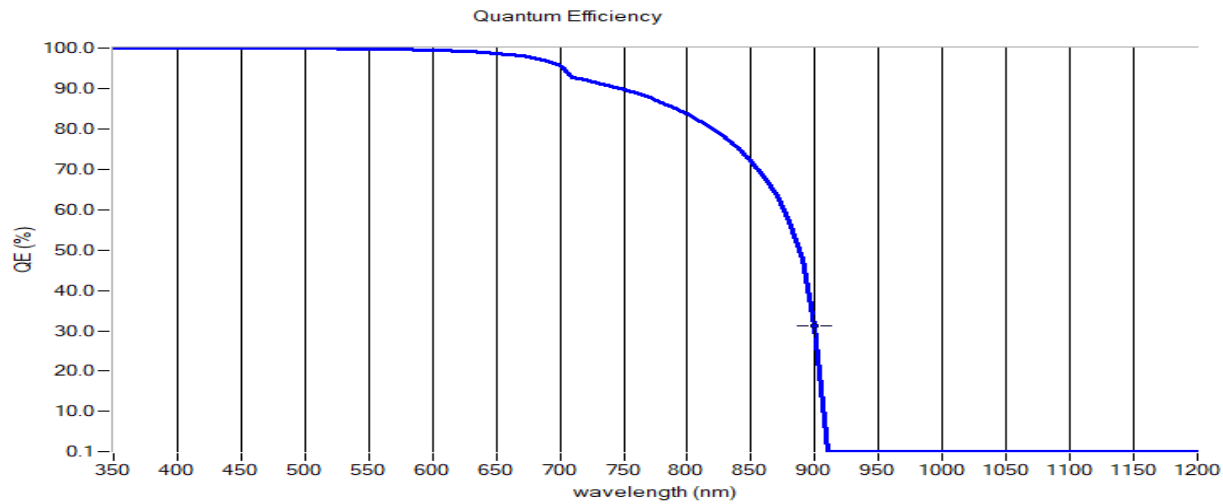
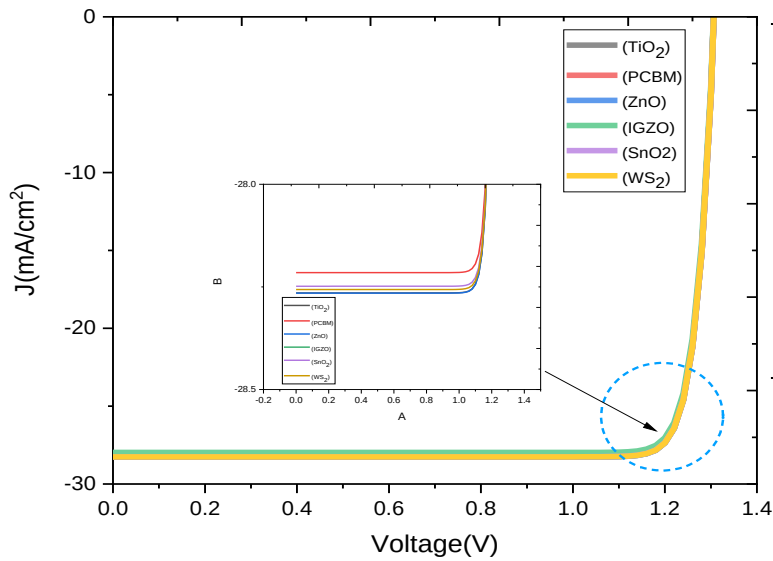


Fig.3 Q-E Graph of bilayer PSC.

Table 5: Experimental results of bilayer PCE.

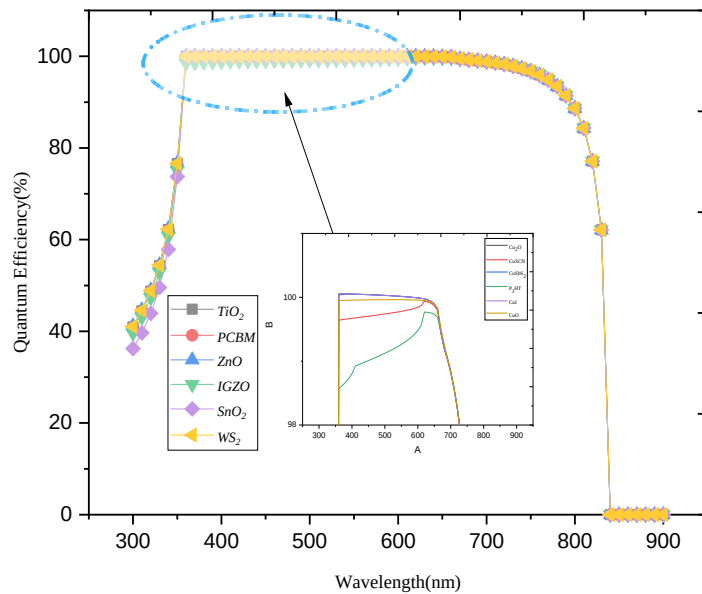
Device Structure	PCE(%)	Voc(V)	Jsc(mAcm ²)	FF(%)	Ref.
ITO/SnS ₂ /CsSnBr ₃ /CsSnI ₂ Br/CBTS/C	30.86	1.25	29.12	84.50	[1]
Calibrated Value	31.71	1.21	29.41	88.91	NA

6.1.2 Comparison of our developed model with simulation results.



The fig. shows how the current-voltage (J-V) performance of trilayer perovskite solar cells varies with different ETL (Electron

Fig.4 J-V Characteristics Graph of trilayer Perovskite cells with different ETL materials.



Displays the quantum efficiency (QE) response for trilayer perovskite cells using various ETL materials, indicating how efficiently light is converted to current.

Fig.5 QE Characteristics of Tri-layer Perovskite cell with different ETL Materials

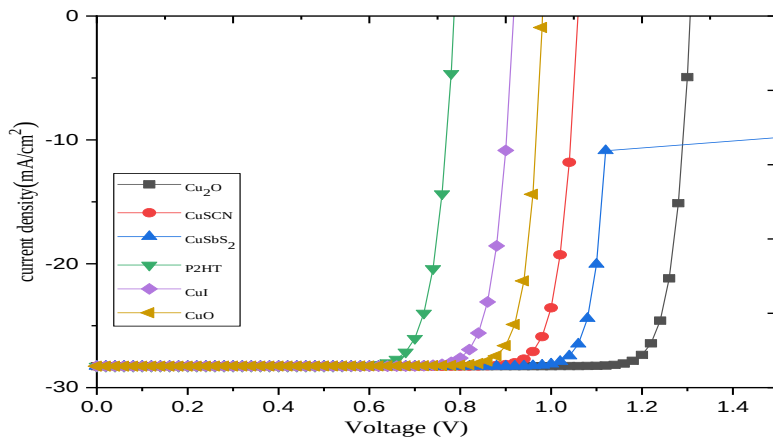


Fig.6 J-V characteristics graph with different HTL materials

Compares the J-V characteristics of trilayer perovskite cells with different HTL (Hole Transport Layer) materials, highlighting their impact on cell efficiency.

6.2: Absorber layer thickness variation

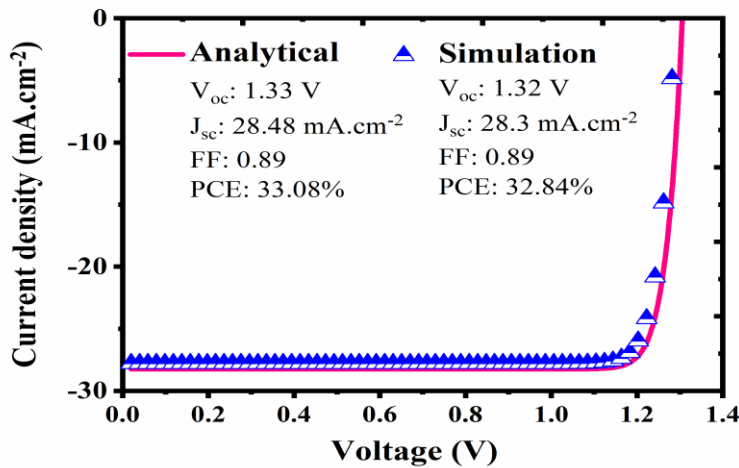


Fig.7 J-V characteristics graph with different ETL materials

Shows calibration results for J-V measurements with different ETL materials to validate the simulation or experimental setup.

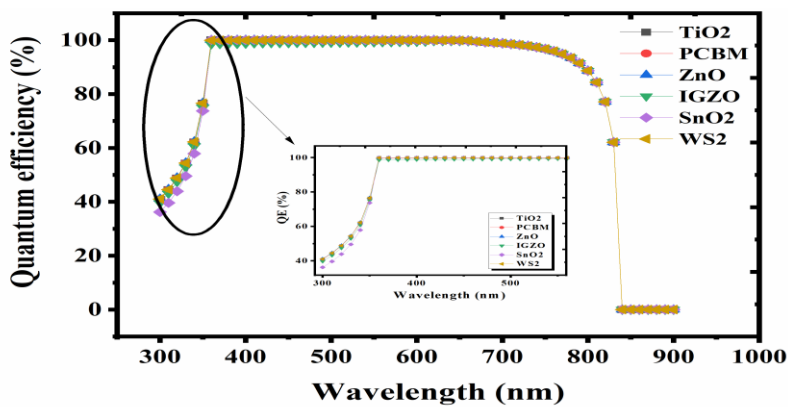
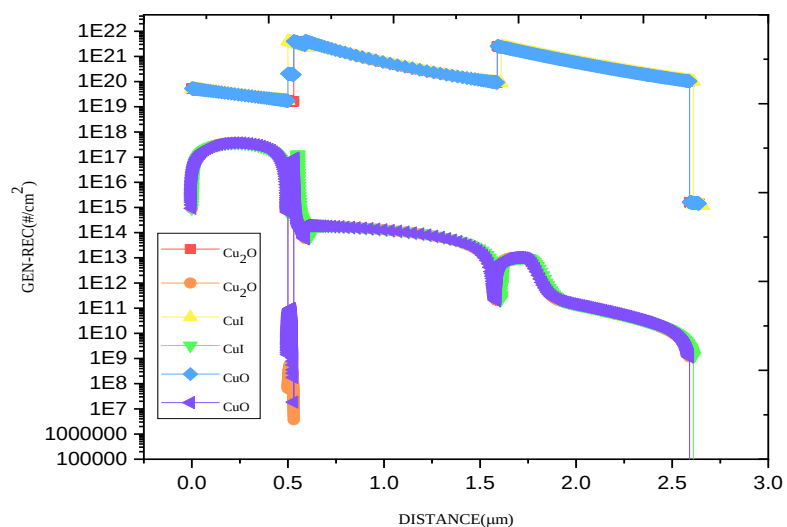


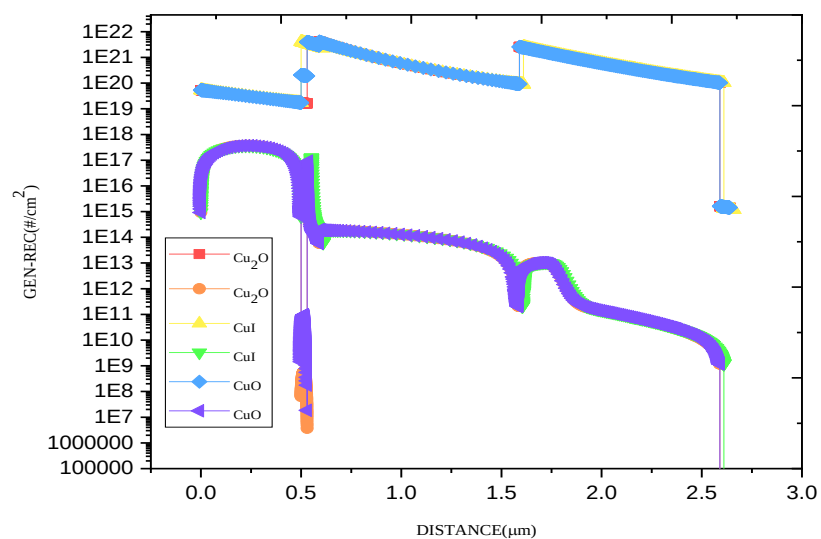
Fig.8 Q-E Graph with different ETL materials.

The fig. shows calibration of the quantum efficiency (Q-E) measurements for different ETL materials to ensure accurate optical modelling.



Illustrates the generation–recombination behaviour in cells with different ETL materials, helping analyze carrier loss mechanisms.

Fig.9 Generation-recombination of different ETL materials.



Illustrates the generation–recombination behaviour in cells using various HTL materials, showing how they affect carrier dynamics.

Fig.10 Generation-recombination of different ETL materials

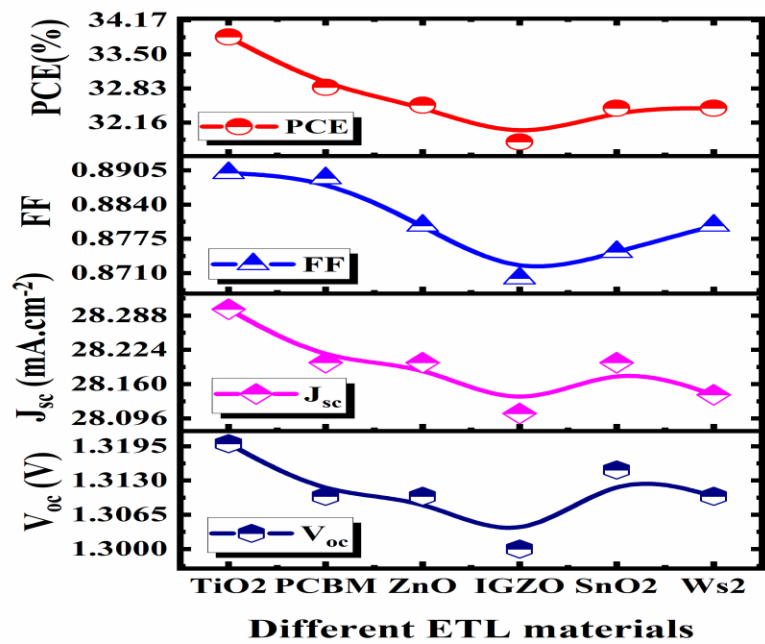
Table 6 : ETL Performance

ETL	Voc (V)	Jsc (mA/cm ²)	FF (%)	Efficiency (%)
TiO ₂	1.30	28.26	88.90	32.84
SnO ₂	1.30	28.24	88.90	32.82
IGZO	1.30	27.97	88.85	32.48

6.3.1: Use of Different ETL Materials

Table 7: Performance matrices of different ETL materials.

Materials	Voc (V)	Jsc (mA/cm ²)	FF(%)	eta(%)
TiO ₂	1.3	28.26	88.90	32.84
PCBM	1.3	28.21	88.90	32.78
ZnO	1.3	28.26	88.90	32.84
IGZO	1.3	27.97	88.85	32.48
SnO ₂	1.3	28.24	88.90	32.82
WS ₂	1.3	28.25	88.90	32.83



Summarizes key performance metrics (like PCE, FF, Jsc, Voc) for different ETL materials in trilayer perovskite cells.

[table 4]

Fig.11 Performance matrices of different ETL materials.

6.3.2: Use of different Materials

Table 8: HTL Performance

HTL	Voc (V)	Jsc (mA/cm ²)	FF (%)	Efficiency (%)
Cu ₂ O	1.30	28.26	88.90	32.84
CuSCN	1.05	28.26	86.95	26.04
P3HT	0.78	28.26	82.90	18.44

Table 9: Performance matrices of different HTL materials

Materials	Voc (V)	Jsc (mA/cm ²)	FF(%)	eta(%)
Cu ₂ O	1.3	28.26	88.90	32.84
CuSCN	1.05	28.26	86.95	26.04
CuSbS ₂	1.13	28.27	89.10	28.54
P ₃ HT	0.78	28.26	82.90	18.44
CuI	0.91	28.26	85.14	22.09
CuO	0.98	28.26	87.10	24.15

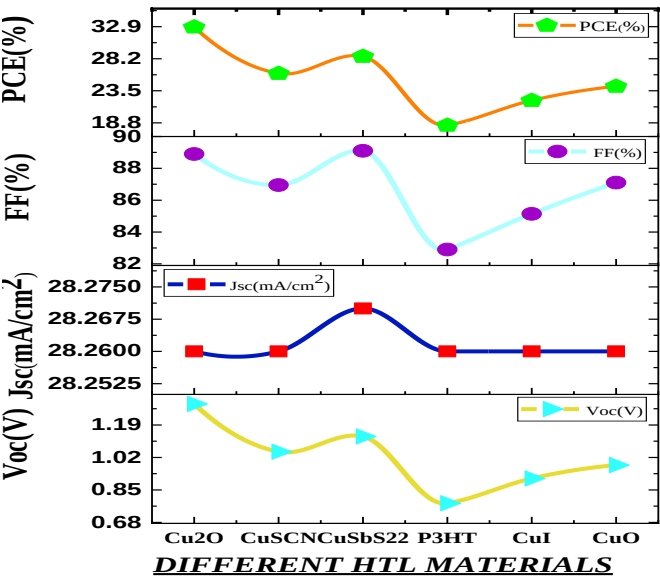


Fig.12 Performance matrices of different ETL material

6.4: Comparative analysis of our results with already reported research materials

Table 10: Simulated and experimental results of trilayer PSCs.

Device Structure	PCE (%)	V _{oc} (V)	J _{sc} (mA/cm ²)	FF (%)	Ref.
ITO/WS2/CsSnBr ₃ /Cu ₂ O/ Au (S)	22.43	1.33	19.59	85.80	[8]
FTO/TiO ₂ /CH ₃ NH ₃ PbI ₃ / CsSnI ₃ /Carbon (S)	21.64	1.08	25.33	79.3	[3]
FTO/TiO ₂ /CsPbI ₃ /FaPbI ₃ / Spiro-OMeTAD/Au (S)	19.94	1.1	24.5	74	[9]
ITO/ SnS ₂ / CsSnBr ₃ /CsSnI ₂ Br / CBTS/C	30.86	1.25	29.12	84.50	[2]
MY WORK	32.84	1.30	28.26	88.90	

EXPERIMENTAL PCE = 30.86%

SIMULATED PCE = 32.84%

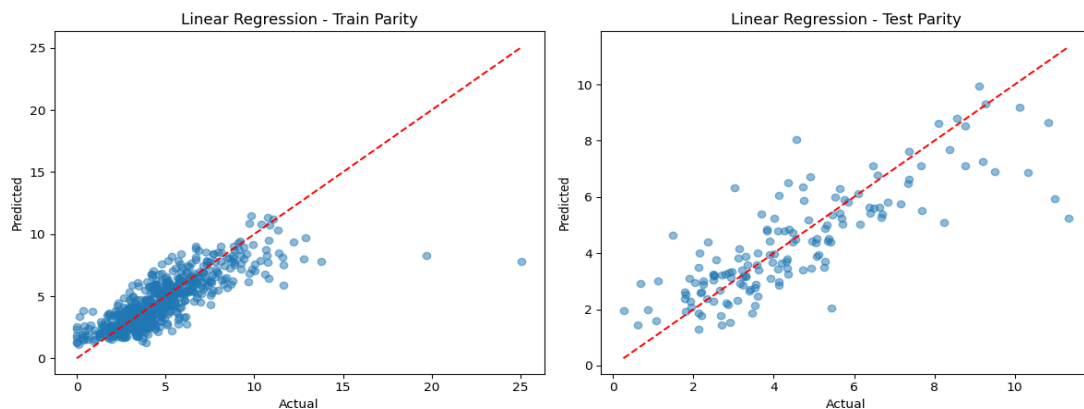
The error due to the constraints in the simulator device = 6.21%

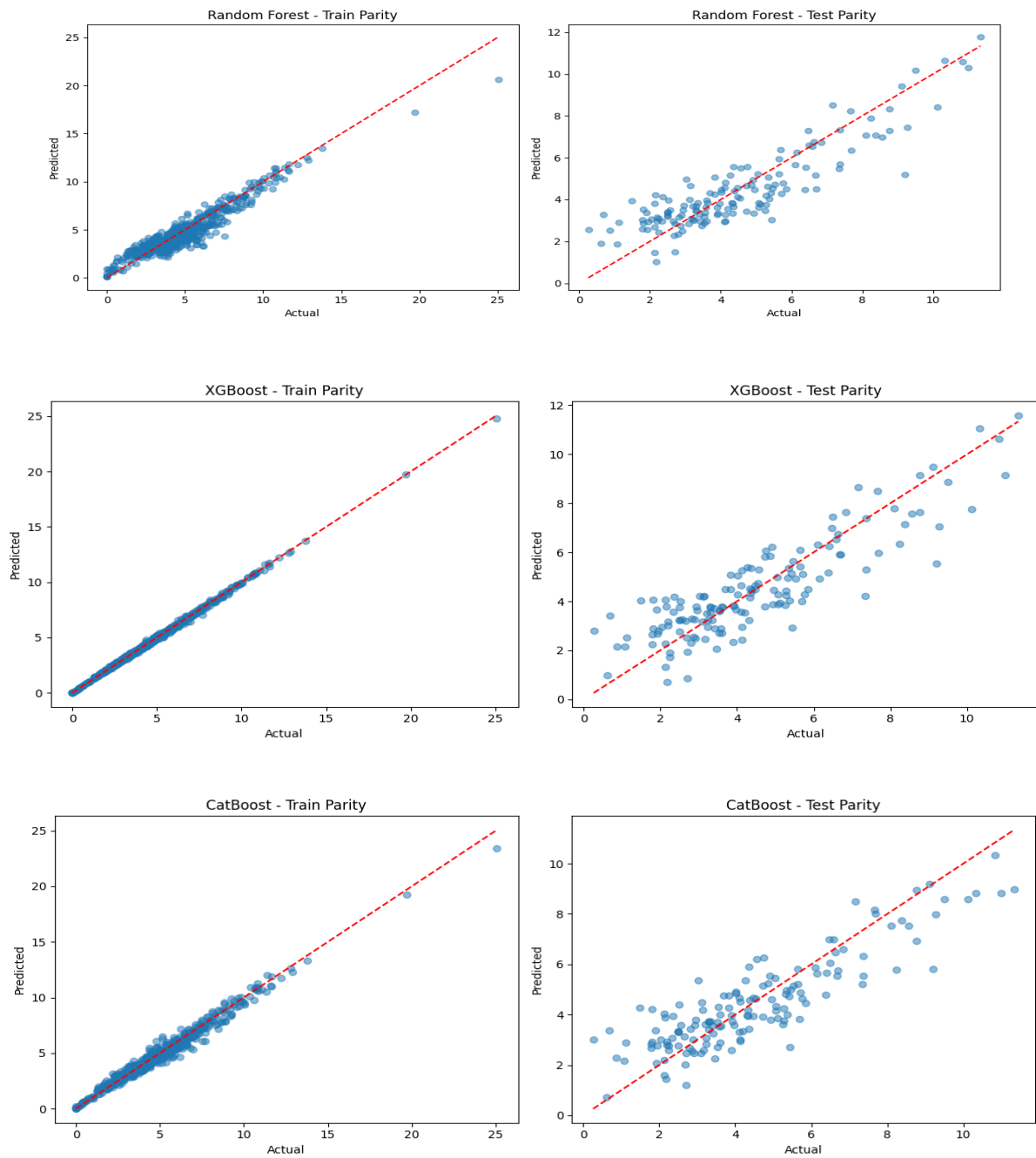
6.5: Machine Learning Modelling for a proposed trilayer Perovskite Solar Cell

OUTPUT 1:

Model Comparison MACHINE:

	Train R ²	Test R ²	Train RMSE	Test RMSE
Linear Regression	0.641863	0.665464	1.571872	1.327507
Random Forest	0.902611	0.783060	0.819685	1.069019
XGBoost	0.998295	0.766293	0.108452	1.109561
CatBoost	0.974849	0.763885	0.416554	1.115262
ANN	0.814524	0.745007	1.131191	1.158989





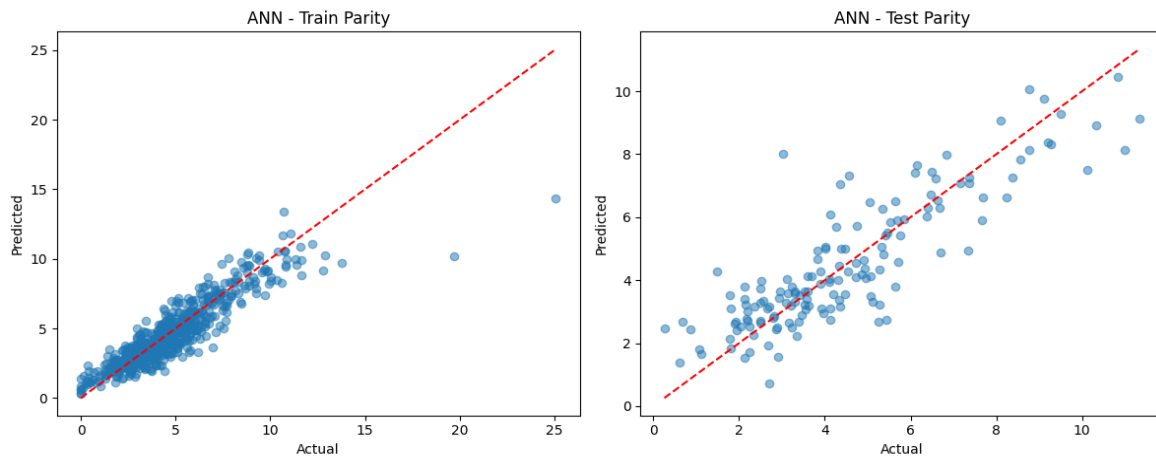


Fig 13 Test and train parity of Linear Regression,ANN,Random Forest,XGBoost,CatBoost

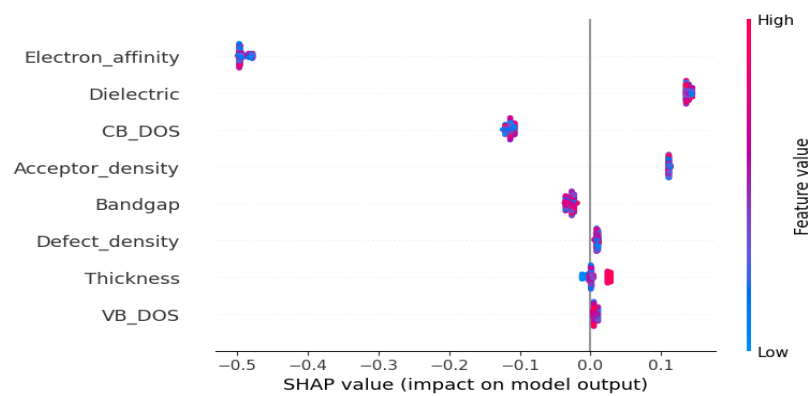


Fig 14: SHAP Value

MACHINE LEARNING MODELLING
OUTPUT 2:

🔍 Predicted PCE from Aggregated Features:
 Linear Regression : 3.797 %
 Random Forest : 20.522 %
 XGBoost : 18.839 %
 CatBoost : 5.378 %
 ANN : 19.463 %

7. Conclusion

- This study presented a thorough investigation into the performance modeling and optimization of a lead-free trilayer perovskite solar cell (PSC) architecture using both analytical methods and machine learning techniques. The proposed device structure, which includes SnS₂ (ETL), CsSnBr₃, CsSnI₂Br (absorbers), and CBTS (HTL), was analyzed using drift-diffusion equations, Poisson's law, and continuity equations under both dark and illuminated conditions. The analytical model effectively captured how generation rate, recombination mechanisms, and electric field distribution affect the current-voltage (J–V) characteristics.
- To speed up performance prediction and find key parameters, five machine learning models were trained on 700 data points generated from simulations. Among these, Random Forest, XGBoost, and Artificial Neural Network (ANN) showed high accuracy. Random Forest provided the best overall performance, achieving a Test R² of 0.783 and a predicted PCE of 20.52%, which closely matched the analytical results.
- Further analysis of materials and interfaces showed that TiO₂ and SnO₂ are highly effective ETLs, while Cu₂O emerged as the most efficient HTL due to its excellent hole mobility and energy alignment. The trilayer absorber setup enhanced light harvesting and carrier collection, leading to simulated efficiencies as high as 32.84%.
- In conclusion, this work confirms the potential of lead-free trilayer PSCs as high-efficiency, environmentally friendly alternatives to lead-based devices. Combining physics-based modeling with machine learning creates a strong framework for speeding up solar cell design and optimization. Future research may focus on experimental validation, interface passivation, and applying this hybrid approach to inverse design and real-time optimization.

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